```
e
(socket: Operation not permitted)
student@fedora:~$ ping google.com
PING google.com (142.250.66.14) 56(84) bytes of data.
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=1
e=29.8 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=2
e=37.9 ms
64 bytes from bom07s35-in-f14.1e100.net (142.250.66.14): icmp_seq=3
=72.0 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=4
e=35.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=5
e=73.9 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=6
e=8.38 ms
64 bytes from pnmaaa-ap-in-f14.1e100.net (142.250.66.14): icmp_seq=7
e=91.9 ms
^C
--- google.com ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6011ms
rtt min/avg/max/mdev = 8.375/49.962/91.899/27.513 ms
```

- mtr :

MTR is a
a dignostic and

- tcpdump :

The tcpdu
and displaying

- ping :

Ping is
to another
Control Message

```
ge: ip [ OPTIONS ] OBJECT {
    ip [ -force ] -batch fil
re  OBJECT := { address | ad

              link | macse

              neighbor | n
              ntbl | route
              token | tunn
   OPTIONS := { -V[ersion]
               -h[uman-rea
               -f[amily] {
               -4 | -6 | -
               -l[oops] { n
               -o[neline]

               -rc[vbuf] [s
               -c[olor]}
```

MATE Terminal

inal Help

```
My traceroute  [v0.95]
t (::1)                                    2026-07-16T1
de   Restart statistics    Order of fields
                    Packets                Pings
                 Loss%   Snt   Last   Avg  Best
                 0.0%    190   0.0    0.1  0.0
```

```
tcpdump
output suppressed, use -v[v
s3, link-type EN10MB (Ether
P fedora.56515 > dns.google
 (53)
P fedora.45469 > dns.google
3)
P dns.google.domain > fedor
et., CNAME mozilla.map.fast
29.91, A 151.101.193.91 (19
P fedora.59512 > dns.google
)
P fedora.48751 > dns.google
(52)
P dns.google.domain > fedor
et., CNAME mozilla.map.fast
, AAAA 2a04:4e42:600::347,
P fedora.60875 > dns.google
```

```
ile  Edit  View  Search  Terminal  Help
udent@fedora:~$ mtr
udent@fedora:~$ su
ssword:
ot@fedora:/home/student# pseudo dnf
sh: pseudo: command not found
ot@fedora:/home/student# sudo dnf
st metadata expiration check: 4:47:1

ckage tcpdump-14:4.99.4-2.fc39.x86_6
pendencies resolved.
thing to do.
mplete!
ot@fedora:/home/student# tcpdump -D
enp0s3 [Up, Running, Connected]
any (Pseudo-device that captures on
lo [Up, Running, Loopback]
bluetooth-monitor (Bluetooth Linux M
usbmon2 (Raw USB traffic, bus number
usbmon1 (Raw USB traffic, bus number
```

RESULT :

Basic
Windows opera
and Verified

Scanned with OKEN Scanner

- **mtr :**

MTR is a program with a CLI that serves as a dignostic and trouble shooting tool

- **tcpdump :**

The tcpdump command is designed for capturing and displaying packets.

- **ping :**

Ping is a tool that verifies IP-Level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP).

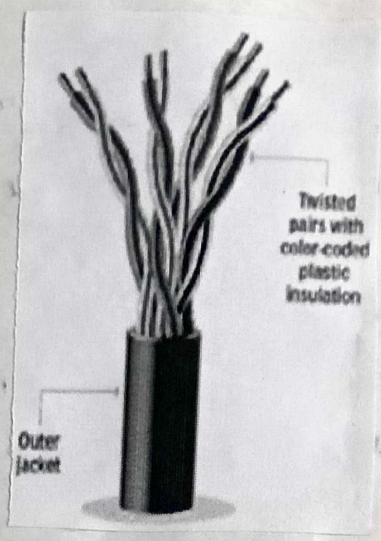
27/7/20

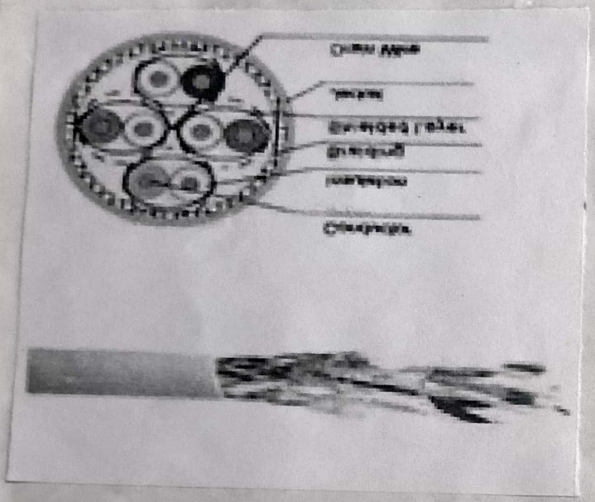
RESULT :

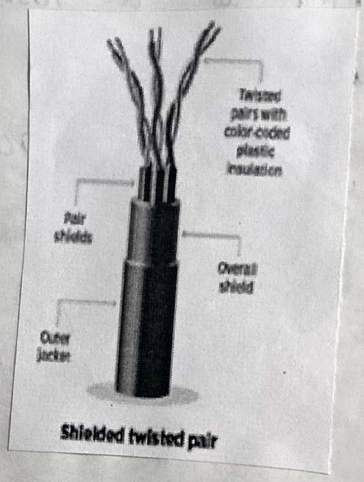
Basic Networking commands in both Linux and Windows operating system is succesfully executed and verified.

Scanned with OKEN Scanner

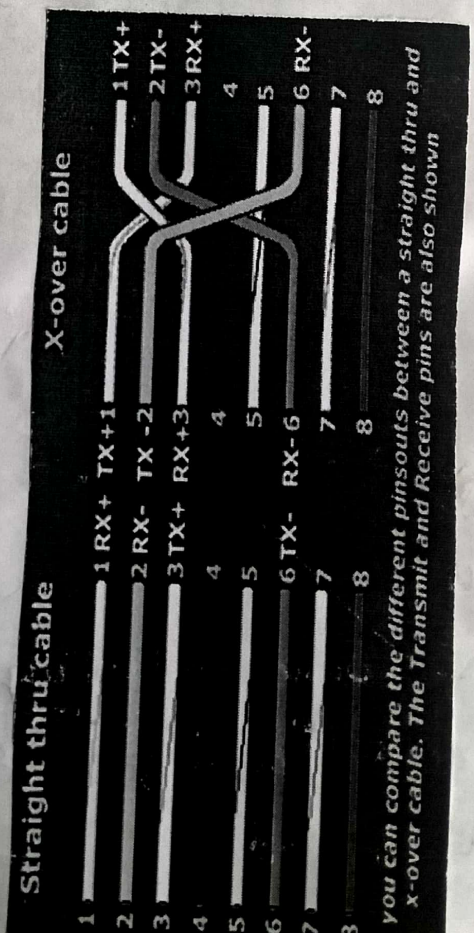












Scanned with OKEN Scanner

23/07/ end

Exp no: 2 STUDY OF DIFFERENT TYPES OF NETWORK CABLE

AIM:

To study and Understand different types of network cable, To make your own ethernet cross over cable / straight cable

DIFFERENT TYPES OF NETWORK CABLE:

1. Unshielded Twisted pair cable
2. Shielded Twisted pair cable
3. Coaxial cable
4. Fibre optic cable.

UTP:

- In Unshielded Twisted pair cable has three categories
- Category 3 has 10bps data Transmission capacity
- Category 5 has upto 100 mbps Data Transmission capacity
- Category 5e has 1Gbps has capacity